MILITARY SPECIFICATION

MULTIPLEXER TD-660( )/G

This amendment forms a part of Military Specification MIL-M-55650B(EL), dated 28 November 1975.

PAGE 1

2.1: Delete "PPP-P-291" and "Paperboard, Wrapping, Cushioning".

Delete "PPP-T-45" and "Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing".

Delete "PPP-T-76" and "Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing)".

Add the following to the bottom of page:

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, N. J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter."

PAGE 2

2:1: Delete "MIL-B-43014" and "Boxes, Water Resistant Paperboard, Folding, Set-up and Metal-Staved".

Delete "MIL-I-46058" and "Insulation Compound, Electrical (For Coating Printed Circuit Assemblies)".

Add (Under STANDARDS) the following:

"FEDERAL

FED-STD-356 Commercial Packaging of Supplies and Equipment".

PAGE 5

3.4.1.2 Line 2: Delete "-4 ±.1db" and substitute "-4 dbm ±0.1 db".

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PAGE 6

3.4.1.6 Line 1: Delete "Radio" and substitute "Ratio".

3.4.1.7 Line 2: Delete "Radio" and substitute "Ratio".

PAGE 8

3.4.5.1 Line 1: Delete "4.8.1" and substitute "4.8.6".

PAGE 13

3.6.1, 3.6.1.1, 3.6.1.2, 3.6.2, 3.6.3, 3.6.4.1, 3.6.4.2, 3.6.4.3 delete and substitute the following:

"3.6.1 Temperature. The following low and high temperature requirements shall be met:

3.6.1.1 Low (see 4.6.1.1). The equipment shall be operated during and after testing as specified in 4.6.1.1, and shall meet the requirements of 4.8.8 at room ambient temperature and at -25°F. The equipment shall withstand exposure to a non-operating temperature of -65°F.

3.6.1.2 High (see 4.6.1.2). During and after testing as specified in 4.6.1.2 the equipment shall operate and meet the requirements of 4.8.8 at room ambient and +125°F with simultaneous solar radiation of 360 BTU/sq. Ft/Hr. The equipment shall withstand exposure to a non-operating temperature of +154°F.

3.6.2 Moisture Resistance (Humidity) (see 4.6.2). During and after testing as specified in 4.6.2 the equipment shall meet the requirements for the measurements as specified in 4.8.7. The equipment shall exhibit no physical damage such as corrosion, rust, blistering swelling or deterioration of parts and materials.

3.6.3 Altitude (Elevation) (see 4.6.3). The equipment shall meet full specification performance of 4.8.7 during and after testing as specified in 4.6.3 at altitudes of 15,000 feet above sea level and shall withstand transportation at altitudes of 50,000 feet above sea level.

3.6.4 Vibration, Bounce, and Shock.

3.6.4.1 Vibration, Resonance Search (see 4.6.4). The equipment shall meet the performance requirements of 4.8.6 during and after testing in accordance with 4.6.4, (frequency range 10 - 55 Hz).

3.6.4.2 Shock Bounce, Vehicular (see 4.6.5). The equipment shall meet the performance requirements of 4.8.6 after being tested in accordance with 4.6.5.

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3.6.4.3 Shock (Bench Handling) (see 4.6.6). The equipment in its servicing condition, except the Card Cell Cover shall be installed, shall withstand shock encountered during servicing and meet the requirements of 4.8.6 when tested in accordance with 4.6.6. There shall be no evidence of physical damage except minor surface abrasion."

PAGE 14

3.6.6 Lines 3 and 4: Delete "or electrical degradation".

3.6.8 Line 1: Delete "Dust (See 4.6.9)" and substitute "Dust (Fine Sand) (see 4.6.9)".

PAGE 15

3.11: Delete and substitute the following:

"3.11 Electromagnetic Compatibility (EMC) (see 4.10). The equipment shall meet the electromagnetic interference emission and susceptibility requirements of MIL-STD-461A, Notice 4, Table A-1, for Class CE equipment except that only tests CE04, CS02, CS06, RE02, RE02.1 and RS03 are applicable. Tests and test methods shall be as specified in MIL-STD-462, Notice 3."

PAGE 16

3.17 Lines 3 and 4: Delete "with a coating material that conforms to MIL-I-46058, type ER (Epoxy)."

3.18 Line 2: Delete "mean-time-between-failure (θo)" and substitute "θo mean-time-between-failure (MTBF)".

PAGE 17

4.2.a Delete "preparation for delivery" and substitute "packaging".

4.2.c(1): Delete "preparation for delivery" and substitute "packaging".

4.2.c(2): Delete "preparation for delivery (see 4.15)" and substitute "packaging (see 4.15)".

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TABLE I - Delete and substitute the following:

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"TABLE I - FIRST ARTICLE INSPECTION (See Notes)

Inspections	Rqt. Para.	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6	Unit 7	Unit 8	Units 9 - 16
1. Inspections covered by subsidiary documents	---	4.4	Inspection to be performed on all units								
2. Bounce preconditioning	3.5.2	4.7.2	Inspection to be performed on all units								
3. Electrical preconditioning	3.5.1	4.7.1	Inspection to be performed on all units								
4. Group A Inspection	(See Table II)		Inspection to be performed on all units								
5. Group B Inspection	(See Table III)		Inspection to be performed on all units								
6. TEMPEST	3.7	4.13	1								
7. Electromagnetic Compatibility (EMC)	3.11	4.10					1				
8. Group C Inspection											
Low Temperature	3.6.1.1	4.6.1.1		1					1		
High Temperature	3.6.1.2	4.6.1.2		2					2		
Moisture Resistance (Humidity)	3.6.2	4.6.2						2			
Altitude (Elevation)	3.6.3	4.6.3			1						
Vibration, Resonance Search	3.6.4.1	4.6.4			2						
Shock Bounce, Vehicular	3.6.4.2	4.6.5			3					2	
Shock (Bench Handling)	3.6.4.3	4.6.6			4						
Fungus (See Note D)	3.6.5	4.6.10				1					
Dripproof	3.6.6	4.6.7						1		1	
Salt Fog (See Note D)	3.6.7	4.6.8		4							
Dust (Fine Sand)	3.6.8	4.6.9		3							
9. Group D Reliability	3.18	4.14									1 (Note c)

NOTES:

a. The inspections 1 through 5, in the order shown, shall be performed on all First Article units before subjecting these units to any other inspection requirements in the table.

b. The numbers in the unit columns in the table covered by Inspections numbered 6, 7, and 8, specify the order of inspections for the indicated unit.

c. Units 9 - 16: These units are to undergo Inspections 1, 2, 3, 4, 5, and 9.

d. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection."

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4.5 Line 1: Delete "Preparation For Delivery." and substitute "Packaging."

4.5.1: Delete and substitute the following:

"4.5.1 Group A Inspection. Each unit (and each subassembly and assembly identified in Table II) on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. The order of inspection of the subassemblies and assemblies and units (end equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in Table II shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in Table II.

b. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table II (Quality Assurance Verification). Factors of lots composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies from the discrete lots which have passed the above Quality Assurance Verification Inspection shall be assembled into the end equipments which shall then be subjected by the contractor to the preconditioning requirements of 3.5 and a 100% visual/mechanical and electrical inspection in accordance with Table II in an order satisfactory to the Government with the Operational Test performed last (See 6.5).

NOTE: Assemblies are not required to undergo further visual/mechanical and electrical inspection as part of the contractor's 100% Group A Inspection of the end equipment.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQL's indicated in Table II. Group A electrical inspection of the end equipment shall be performed in accordance with Table II in an order satisfactory to the Government with the Operational Test Performed last (see 6.5).

NOTE: Assemblies and subassemblies are not required to undergo further visual/mechanical and electrical inspection as part of the above contractor's sampling inspection of the end equipment.

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"TABLE II, Under Operational Test delete the following:

Inspection	Req Para	Ref Fig	Insp Para	AQL	
				MAJOR	MINOR
Signal-to-Noise Ratio	3.4.1.6	16	4.8.5	1.0%	**
Signal-to-Total Distortion	3.4.1.8	16	4.8.5	1.0%	**
Transmit Timing	3.4.5.1	16	4.8.5	1.0%	**
Metering	3.4.7	16	4.8.5	1.0%	**
Channel Monitoring	3.4.9	16	4.8.5	1.0%	**
DC Output	3.4.11.2	16	4.8.5	1.0%	**"

PAGE 22

4.5.1.1: Delete in its entirety.

PAGE 23

TABLE IV, Delete and substitute the following:

"TABLE IV - GROUP C INSPECTION

Inspection	Req Para	Ref Fig	Insp Para
Sub-Group I			
Temperature:			
Low	3.6.1.1	32	4.6.1.1, 4.8.8
High	3.6.1.2	32	4.6.1.2, 4.8.8
Moisture Resistance (Humidity)	3.6.2	32	4.6.2, 4.8.7
Sub-Group II			
TEMPEST	3.7		4.13
Sub-Group III			
Electromagnetic Compatibility (EMC)	3.11	24	4.10
Dripproof	3.6.6	32	4.6.7, 4.8.7
Shock (Bench Handling)	3.6.4.3	33	4.6.6, 4.8.6
Vibration, Resonance Search	3.6.4.1	33	4.6.4, 4.8.6
Shock Bounce, Vehicular	3.6.4.2	33	4.6.5, 4.8.6
Sub-Group IV			
Altitude (Elevation)	3.6.3	32	4.6.3, 4.8.7
Dust (Fine Sand)	3.6.8	33	4.6.9, 4.8.6 (Note 1)
Salt Fog	3.6.7		4.8.6 (Note 1)
Fungus	3.6.5		4.8.10 (Note 1)

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried and refurbished if necessary, before proceeding with subsequent tests."

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4.5.3.1.1, Line 4: Delete "temperature and humidity" and substitute "Temperature and Moisture Resistance (Humidity)."

4.5.3.1.3, Delete last sentence and substitute "One unit shall be subjected to EMC, Dripproof and Shock (Bench Handling) and one unit shall be subjected to Vibration, Resonance Search, and Shock Bounce, Vehicular with each unit subjected to testing in that order."

4.5.3.1.4, Line 4: Delete "Dust" and substitute "Dust (Fine Sand)".
Line 5: Delete "fungus" and substitute "Fungus"

4.5.3.3: Delete and substitute with:

"4.5.3.3 Group C Failure. (see 6.). Actions required relative to Group C failures shall be as as specified in the contract or purchase order."

PAGE 25

4.5.5 Delete the last sentence.

4.6 After "Service Condition tests" add "(see 3.6)."

PAGE 26

4.6.1.2, Line 7: Delete "125° +5° -0°F" and replace with "125°F +5°F -0°F".

4.6.2 Delete "Humidity (Moisture-Resistance)" and replace with "Moisture-Resistance (Humidity)".

4.6.3 Delete "Low pressure (Altitude) (see 3.6.3)" and substitute, "Altitude (Elevation) (see 3.6.3)."

PAGE 27

4.6.3.c Delete and substitute the following:

"c. Increase the chamber pressure to 429.1mm of Hg (16.9 in of Hg or 15,000 feet above sea level) and then perform the test of 4.8.7 and record the results."

4.6.4 Delete "Vibration Test (Resonance Search) (see 3.6.4.1)" and substitute "Vibration, Resonance Search (see 3.6.4.1)."

4.6.4.d, Lines 2, 4, and 7 Delete "cps" and substitute "Hz".

PAGE 28

4.6.7.b, Line 2, Delete "Figure 33" and substitute "Figure 32".

4.6.9 Delete "Dust Test" and substitute "Dust (Fine Sand)".

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PAGE 29

4.6.10 Delete the first sentence and substitute: "The equipment in its normal operating configuration shall be subjected to the Fungus Test specified in Method 508.1, Procedure I of MIL-STD-810."

PAGE 30

4.7.1.1.b, Last sentence: Delete "final electrical assembly tests by the contractor." and substitute "submission to subsequent quality conformance inspections."

PAGE 31

4.8.2.c, Line 3: Delete "TM 11-5805-383-12" and substitute "TM 11-5805-382-12".

PAGE 34

4.8.6, Line 1: Delete "System Test (service condition/non-operating)." and substitute "Service Condition Test (Vibration, Resonance Search; Shock Bounce, Vehicular; Shock (Bench Handling); and Dust (Fine Sand))".

PAGE 35

4.8.7, Line 1: Delete "System test (service condition/operating)." and substitute "Service Condition Test (Moisture Resistance (Humidity); Dripproof; and Altitude (Elevation))."

4.8.8 Line 1: Delete "System Test (service condition temperature)." and substitute "Service Condition Test (Temperature)."

PAGE 49

4.10 Delete and substitute the following:

"4.10 Electromagnetic Compatibility (see 3.11). The equipment shall be tested for compliance to the requirements of MIL-STD-461, Notice 4, Table A-1, for Class CE equipment. The test methods and procedures described in MIL-STD-462, Notice 3 shall be employed. All electromagnetic compatibility tests shall be performed with the equipment simulating normal operation. The frequency range of the following tests are modified as follows: RE02 -14 khz to 1000 Mhz and RS03 -14 Khz to 1000 Mhz.

a. Connect equipment per figure 24.

b. Align equipment for 12 channel operation.

c. Perform radiated and conducted emission tests on the equipment in accordance with MIL-STD-462, Notice 3 and the approved EMC/EMI test plan.

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d. Perform radiated and conducted susceptibility tests on the equipment in accordance with MIL-STD-462, Notice 3 and the approved EMC/EMI test plan while monitoring the following:

(1) Signal-to-Noise Ratio on one audio channel.

(2) Signal-to-Total Distortion Ratio on one audio channel.

(3) Listen on one audio channel.

e. The requirements of 3.11 shall be met."

PAGE 50

4.14.1.3 Delete and substitute the following:

"4.14.1.3 The detailed Test Conditions, Procedures and Methods for the test described in 4.14.1 and 4.14.2 shall be in accordance with the approved Reliability Test Plan for Multiplexer TD-660( )/G."

PAGE 51

4.15 Delete and substitute the following:

"4.15 Quality Conformance Inspection of Packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5."

5. Delete in its entirety and substitute the following:

"5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or commercial as specified (see 6.2.b).

5.1.1 Level A.

5.1.1.1 Cleaning. Multiplexer TD-660( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Multiplexer TD-660( )/G shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

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5.1.1.4.2 Multiplexer TD-660( )/G. Each Multiplexer TD-660( )/G shall be individually unit packed method III as follows: Secure the tool adjustment within the snap fastener provided on the unit. Cushion the item on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. The cushioned item shall be inclosed within a close fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1 on top of the contents, directly under the lid of the box. Box closure shall be in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial Preservation. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be A, B or commercial, as specified (see 6.2.b). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Multiplexer TD-660( )/G, packaged as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, types 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.2 Level B. A quantity of Multiplexer TD-660( )/G unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, domestic type; PPP-B-621, style 4, class 1, or PPP-B-585, types 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.2.3 Commercial packing. Packing shall be in accordance with FED-STD-356.

5.3 Marking.

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5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356."

PAGE 54

6.2.b Delete "Levels of preservation, packaging" and substitute "Level A, Level B or commercial preservation."

PAGE 55

6.4 Delete in its entirety.

PAGE 56

6.9 Add as follows:

"6.9 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods."

PAGE 75

Figure 32 - Title

Delete and substitute the following:

"Service Condition Test (Temperature; Moisture Resistance (Humidity); Dripproof; and Altitude (Elevation))."

PAGE 76

Figure 33 - Title

Delete and substitute the following:

"Service Condition Test (Vibration, Resonance Search; Shock Bounce, Vehicular; Shock (Bench Handling); and Dust (Fine Sand))."

Custodian:
Army - EL

Preparing Activity:
Army - EL

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